Hitesh Rasineni

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ABOUT ME

Computer Science undergraduate focused on machine learning systems and scientific computing, with hands-on experience building cloud-native ML orchestration platforms using Docker, Kubernetes, MLflow, and observability tools. Research experience in neural density estimation and anomaly detection on collider data, alongside backend engineering, experiment tracking, and reproducible ML workflows.

EDUCATION

VIT-AP University, Amaravati, Andhra Pradesh, India **September 2023 – September 2027**
B.Tech, CSE (AI & ML) CGPA: 8.05 / 10

Sree Vidyanikethan International School, Tirupati, India **2023**
CBSE (10+2) Grade: 91.6 / 100

SELECTED PROJECTS

Cloud-Native ML Training Platform *FastAPI, Docker, Kubernetes, Next.js, PyTorch, MLflow*
Code: github.com/HiteshRasineni/Cloud-Native-ML-Training-Platform-for-Scientific-Data

- Built a cloud-native ML experiment-orchestration platform using FastAPI, PostgreSQL, Redis, Docker, and Kubernetes, with queued job scheduling, lifecycle tracking, retries, heartbeats, and pluggable local and Kubernetes executors.
- Implemented experiment workers and MLflow tracking with MinIO-backed artifacts, plus Prometheus/Grafana observability, a Next.js interface for experiment submission and loss-curve visualization, and pytest suites for backend, scheduler, and worker components.
- Added a dataset registry with versioned MinIO storage and a workload-agnostic interface running a from-scratch RealNVP normalizing flow in PyTorch.

ExperimentIQ — Query-Routing RAG over MLflow Experiments *FastAPI, MLflow, Qdrant, sentence-transformers*
Code: github.com/HiteshRasineni/ExperimentIQ

- Developed a containerized RAG assistant for MLflow experiments, routing structured metric and parameter queries to the tracking API and unstructured questions to Qdrant vector retrieval.
- Implemented source-grounded responses with run-level references and confidence scores, and evaluated routing and answer generation on an initial hand-authored 10-question benchmark.
- Shipped via docker-compose (Qdrant + API) with seed scripts and an eval harness reporting per-category results.

The Unlikely Lab — Personal Research Publication Platform *Next.js, TypeScript, Pandoc*
Code: github.com/The-Unlikely-Lab/the-unlikely-lab

- Designed, built, and deployed a research-publication platform hosting complete typeset HTML versions of arXiv-style physics papers plus an MDX content system, with an automated Pandoc LaTeX-to-HTML pipeline (MathML/KaTeX, figure/table validation) as a statically exported Next.js app on GitHub Pages via CI/CD.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript / TypeScript (React, Next.js)

ML/Data: PyTorch, scikit-learn, OpenCV, Pandas, Matplotlib, Density Estimation (Normalizing Flows, Likelihood Ratios), RAG

Web/Platform: Next.js, React, FastAPI, TypeScript, Tailwind CSS, MDX

MLOps & Infra: MLflow, Qdrant, Docker, Kubernetes, PostgreSQL, Redis, MinIO, Prometheus, Grafana, GitHub Actions (CI/CD)

Tools: Git, Pandoc, LaTeX, uproot, Google Colab

RESEARCH EXPERIENCE

Dark Matter Mono-Z (Density Estimation)

Leptonic Mono-Z Dark Matter Search with Neural Spline Flows
H. Rasineni, B. Chebrolu. arXiv preprint, 2026 (submitted for journal publication). *Leptonic decay channel.*
DOI: 10.48550/arXiv.2607.13771
Code: github.com/HiteshRasineni/Leptonic-Mono-z-CMS2015-DarkMatter-Search

– Trained five neural spline flows on 2.32 fb^{-1} of CMS Run 2015D data ($\mu\mu/ee$ channels, 37-dim features) and set 95% CL signal-strength limits via an SR+VR profile-likelihood fit: $\mu < 0.018$ (scalar), 0.036 (vector), 0.050 (axial-vector).

Hadronic Mono-Z Dark Matter Sensitivity with Flow Matching
H. Rasineni, B. Chebrolu. arXiv preprint, 2026. *Hadronic decay channel.*
DOI: 10.48550/arXiv.2609.02923
Code: github.com/HiteshRasineni/CMS2015DarkMatterSearch-HTMHT-

– Built the MadGraph5_aMC@NLO → Pythia 8 → Delphes pipeline (1.44M selected events) with a conditional flow-matching background model; projected expected significances of 2.89σ, 7.62σ, and 7.41σ, with extra-jet kinematics carrying 53–71% of the discriminating power.

Likelihood-Based Unsupervised Anomaly Detection in CMS Dijet Events
B. Chebrolu, H. Rasineni, P.A. Immadi (equal contribution). arXiv preprint, 2026.
DOI: 10.48550/arXiv.2609.06686
Paper: the-unlikely-lab.github.io/the-unlikely-lab/research/dijet-anomaly

- Unsupervised, model-independent anomaly search on 378,278 CMS Run 2011B HT events ($\sqrt{s} = 7$ TeV): a neural spline flow learned the SM multijet background from jet-substructure and event-shape observables; the 99.9th-percentile low-likelihood tail was validated with mass-decorrelation, permutation-null, and stability tests (overlap $> 88\%$ across seeds, rank correlation > 0.95).

Improving Discovery-Significance Stability in Higgs Event Classification
J.J. Pujari, P.A. Immadi, H. Rasineni, T. Bikku, R.S. Puppala. *Discover Artificial Intelligence* (Springer Nature), 2026.
DOI: 10.1007/s44163-026-01683-5

- Supervised contrastive pre-training with a parallel FT-Transformer + XGBoost ensemble reduced fold-to-fold variance in Approximate Median Significance (AMS = 3.74) versus focal-loss training, enabling more stable threshold selection.

CERTIFICATIONS

- **Microsoft Certified: Azure AI Fundamentals** (Credential)
- **AWS Academy Graduate – Cloud Foundations (Training Badge)** (Credential)
- **WorldQuant University, Applied AI Lab – Deep Learning for Computer Vision** (YOLOv8, MTCNN, Inception-ResNetV1) (Credential)
- **IBM Machine Learning Specialist (Professional V1)** (EDA, ensembles, transfer learning) (Credential)
- **NVIDIA DLI – Fundamentals of Deep Learning** (Credential)
- **NVIDIA DLI – Transformer-based NLP** (Credential)